Status	Finished
Started	Tuesday, 20 October 2026, 5:50 PM
Completed	Tuesday, 20 October 2026, 5:55 PM
Duration	5 mins 7 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The search continues in the left or right sublist ✓
- b. The list is sorted
- c. The search continues from the beginning
- d. The search stops

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

- a. Binary search
- b. Dynamic search
- c. Linear search ✓
- d. Hashing

Question **3**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

- a. At the beginning of the list
- b. In the right sublist
- c. In the left sublist ✓
- d. In the entire list

Question **4**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

- a. Deleting elements from a list
- b. Sorting elements in a list
- c. Inserting elements into a list
- d. Determining whether an element is present in a list ✓

Question **5**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. n/2
- b. log n
- c. n-1
- d. n ✓

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

- a. Skipping every second element
- b. Dividing the list in half repeatedly
- c. Checking each element sequentially ✓
- d. Sorting the list before searching

Question **7**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

- a. n/2
- b. log n
- c. n ✓
- d. 1

Question **8**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- a. O(1)
- b. O👎 ✓
- c. O(log n)
- d. O(n log n)

Question **9**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

- a. By sorting the list first
- b. By dividing the list into halves
- c. By comparing each element in the list sequentially ✓
- d. By using a hash function

Question **10**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

- a. When the item is somewhere in the middle of the array ✓
- b. When the item is not the array at all
- c. When the item is the last element in the array
- d. Item is the last element in the array or item is not there at all

Question **11**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

- a. It can be applied only if the list is sorted ✓
- b. It always starts from the beginning of the list
- c. It works on unsorted lists
- d. It compares elements sequentially

Question **12**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By starting from the first element
b. By comparing each element sequentially
c. By dividing the list length by two ✓
d. By using a hash function

Question **13**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Both (1) & (3)
b. Binary search
c. Hash search
d. Linear search ✓

Question **14**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search does not require dividing the list
b. Binary search works on unsorted lists
c. Binary search can find multiple instances of the target element
d. Binary search has a lower time complexity on large, sorted lists ✓

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

- a. Binary search algorithm is not efficient when the data elements more than 1500 ✓
- b. There must be a mechanism to access middle element directly
- c. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
- d. Must use a sorted array